LAB 6 REPORT

Constrained Dynamics. Nonholonomic constraints

Team members: Anas Khmais Hamrouni

1. Introduction

The goal of today's lab is to simulate a double pendulum (again:)), but using constrained dynamics and practice on deriving nonholonomic constraints. Also we will practice the same approach on the boat and differential drive systems.

Fill in this report and **submit it to Moodle**.

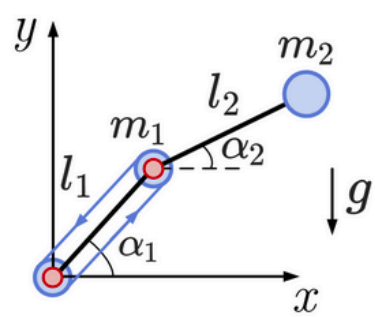
2. Simulation

<u>Tasks</u>:

2.1. Derive the dynamics of the double pendulum

- Write the free body motion of each points
- Set the holonomic constraints and obtain the Jacobians
- Set the nonholonomic constraints and obtain the Jacobians
- Plug in all necessary matrices to the equation of motion subject to constraints

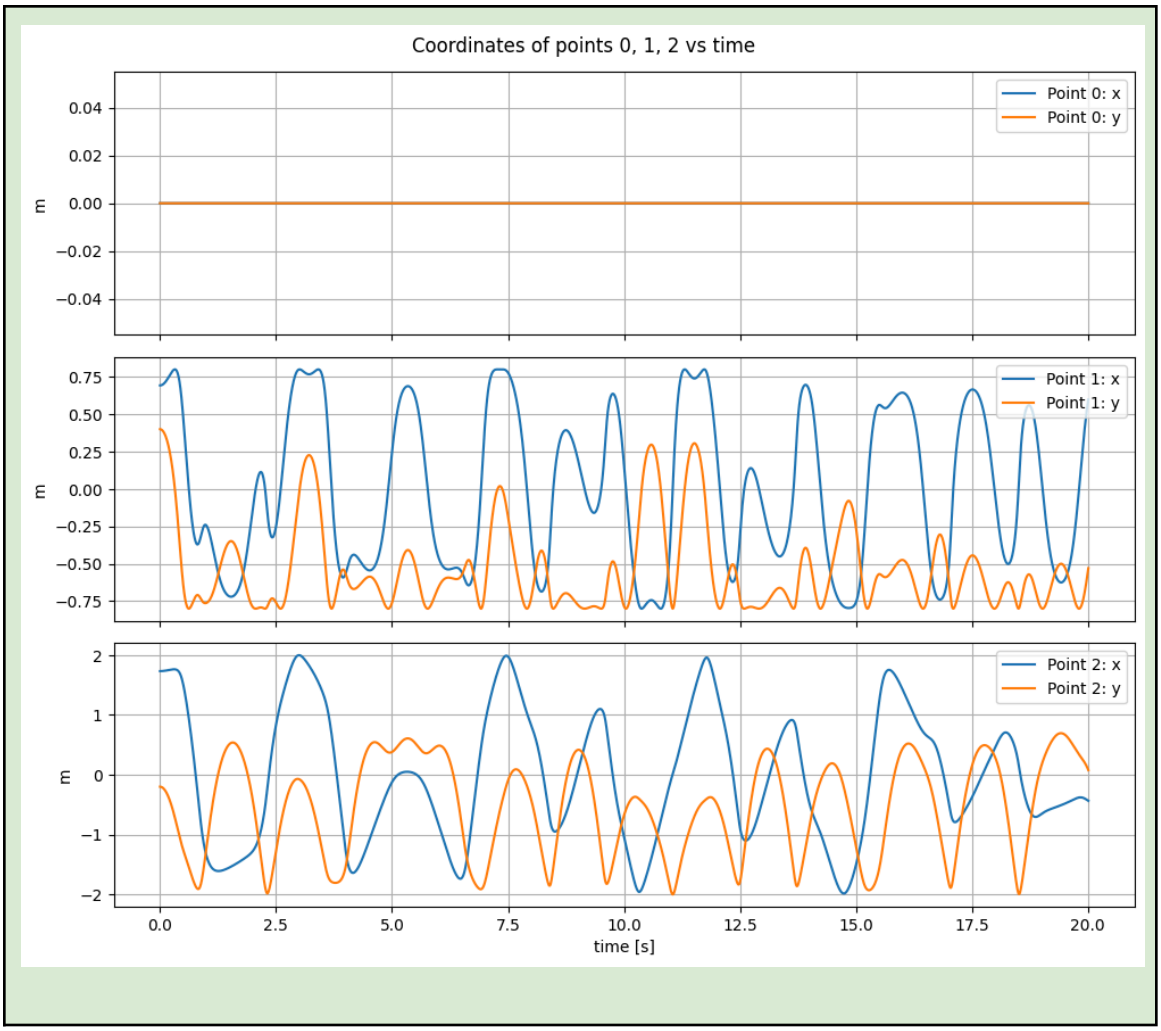
$m_1 = 1$kg
$m_2 = 1$kg
$l_1 = 0.8$ m
$l_2 = 1.2$ m
$g = 9.8$ m/s^2

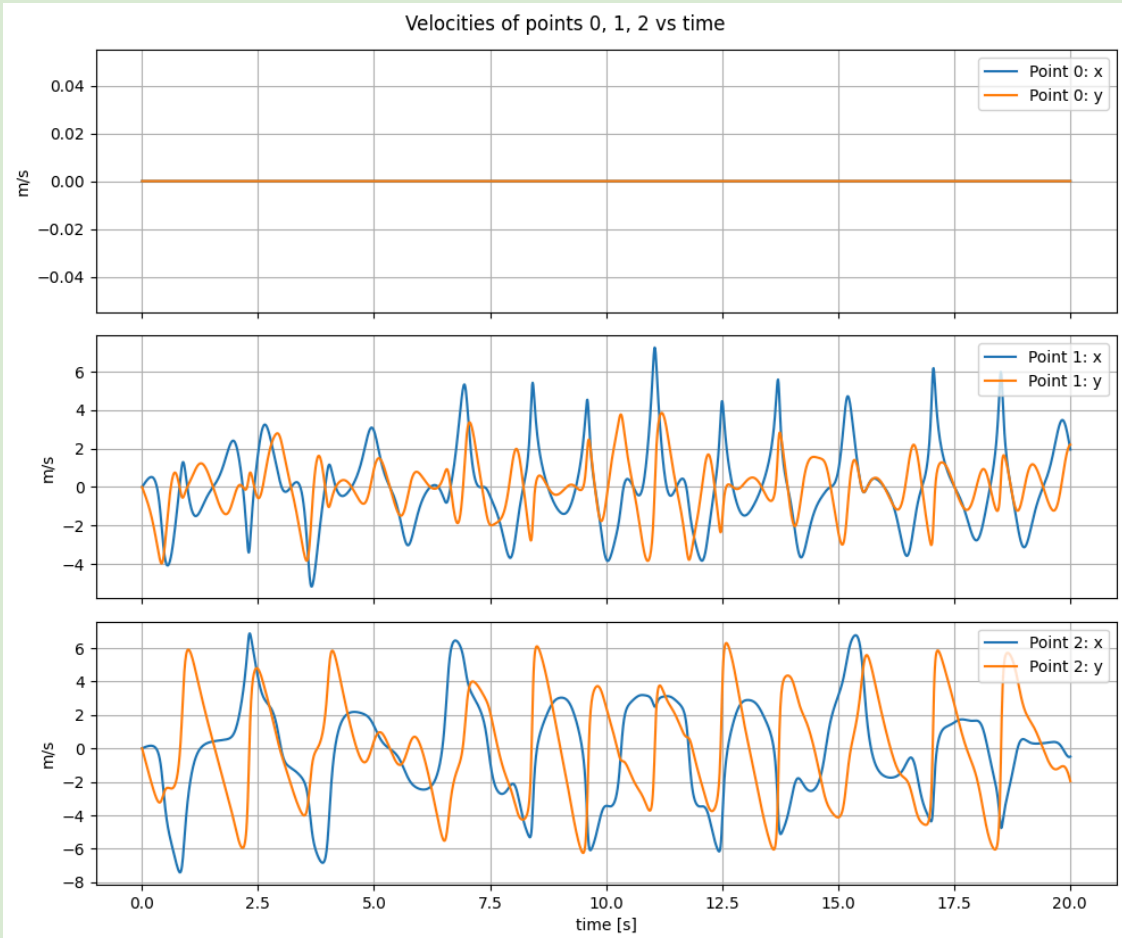


2.2. Simulate the double-link pendulum and visualize the motion (use visualize.py from lab 4). Display in the animation the lengths of each link. Plot the coordinates of the points 0, 1 and 2 versus time and velocities of the points 0, 1 and 2 versus time.

Note: Add a point 0 to visualize.py, this will help you to solve the next tasks.

Notebook + simulation + dynamics derivations


Coordinates of points 0, 1, 2 vs time
Point 0: x
Point 0: y
Point 1: x
Point 1: y
Point 2: x
Point 2: y
m
time [s]
0.04
0.02
0.00
−0.02
−0.04
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
2
1
0
−1
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



Plots for Task 2.2

Explain the obtained result.

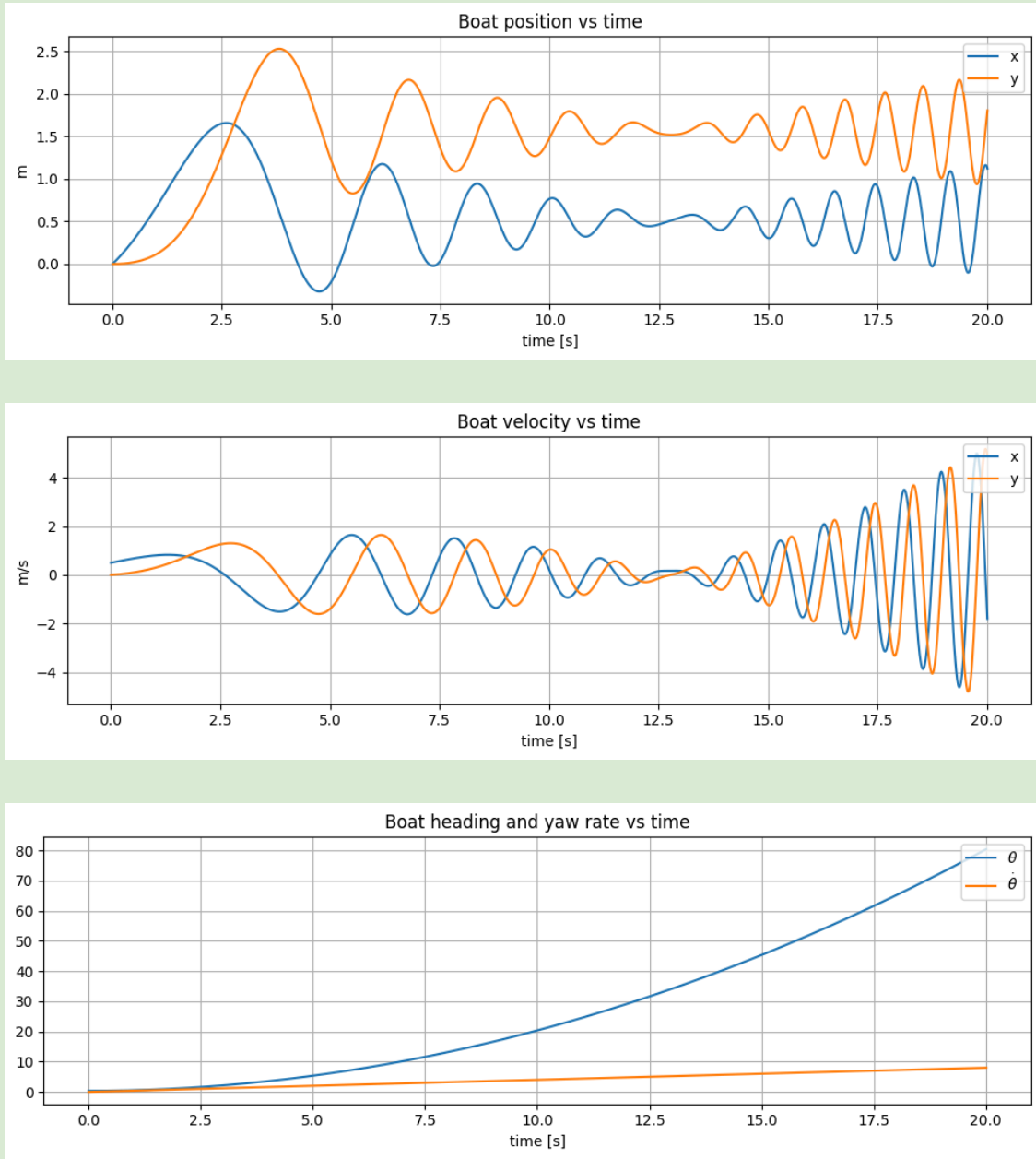
Answer(s):

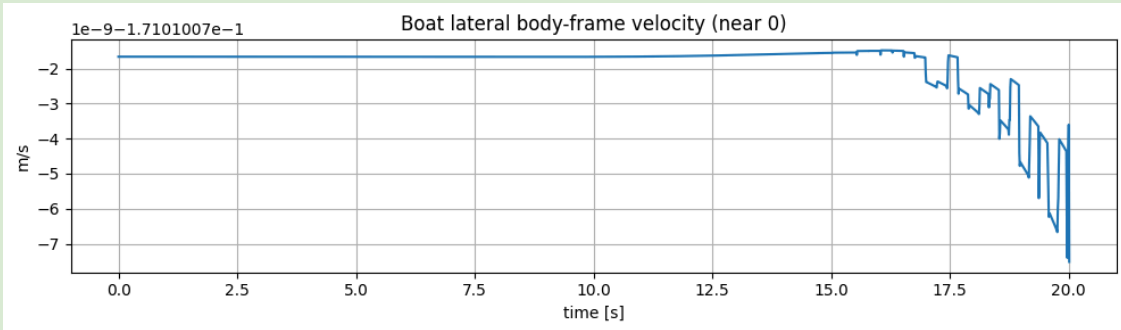
Point 0 stays fixed at the origin, so its coordinates and velocities are identically zero.

Points 1 and 2 oscillate, with point 2 showing larger excursions and higher velocity peaks because its motion combines both links' rotations (due to energy transfer).

The irregular, non-repeating pattern is expected from the nonlinear double-pendulum dynamics, while L1,L2 remaining constant confirms the rigid-link constraints are respected.

2.3. Perform the same approach to the boat model, which you considered in the lecture.





Plots for Task 2.3

How did you set the constraints?

Answer(s):

The boat model assumes **no lateral slip** (zero body-frame lateral velocity):

$$\psi(\boldsymbol{q}, \dot{\boldsymbol{q}}) = -\sin\theta\,\dot{x} + \cos\theta\,\dot{y} = 0.$$

$\boldsymbol{A}$ and b

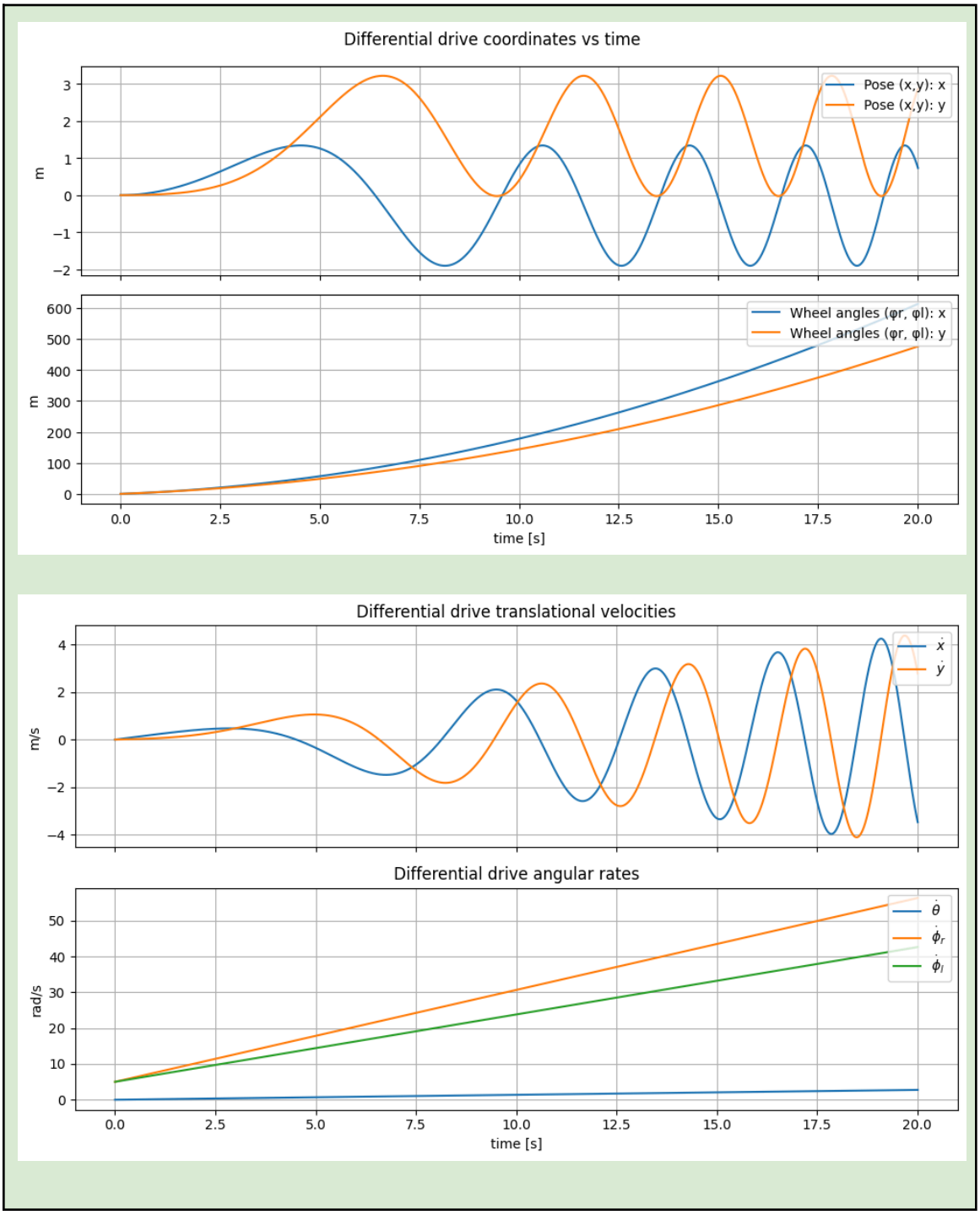
$$\boldsymbol{A}(\boldsymbol{q}) = \frac{\partial\psi}{\partial\dot{\boldsymbol{q}}} = \begin{bmatrix} -\sin\theta & \cos\theta & 0 \end{bmatrix}.$$

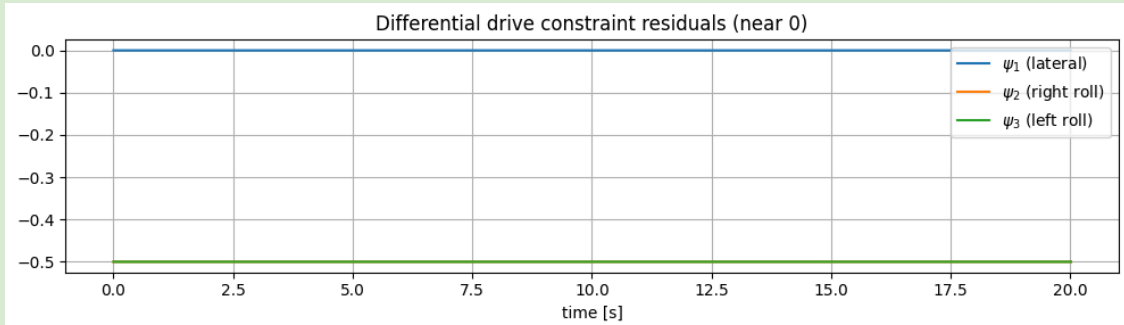
Since $\partial\psi/\partial t = 0$ and only θ appears in ψ,

$$b(\boldsymbol{q}, \dot{\boldsymbol{q}}) = -\frac{\partial\psi}{\partial\theta}\dot{\theta} = \big(\cos\theta\,\dot{x} + \sin\theta\,\dot{y}\big)\dot{\theta}.$$

Thus, the constraint is $\boldsymbol{A}(\boldsymbol{q})\ddot{\boldsymbol{q}} = b(\boldsymbol{q}, \dot{\boldsymbol{q}})$.

2.4. Perform the same approach to the differential drive model, which you considered in the lecture.





Plots for Task 2.4

How did you set the constraints?

Answer(s):

Using the differential drive model:

- **No lateral slip**:
$$\psi_1 = -\sin\theta\,\dot{x} + \cos\theta\,\dot{y} = 0.$$
- **Pure rolling** (no longitudinal slip) at each wheel:
$$\psi_2 = \dot{x}\cos\theta + \dot{y}\sin\theta + \frac{L}{2}\dot{\theta} - r\dot{\phi}_r = 0, \qquad \psi_3 = \dot{x}\cos\theta + \dot{y}\sin\theta - \frac{L}{2}\dot{\theta} - r\dot{\phi}_l = 0.$$

$\boldsymbol{A}$ matrix

Stack $\boldsymbol{\psi} = [\psi_1, \psi_2, \psi_3]^\mathsf{T}$, then $\boldsymbol{A} = \partial\boldsymbol{\psi}/\partial\dot{\boldsymbol{q}}$:

$$\boldsymbol{A}(\boldsymbol{q}) = \begin{bmatrix} -\sin\theta & \cos\theta & 0 & 0 & 0 \\ \cos\theta & \sin\theta & \frac{L}{2} & -r & 0 \\ \cos\theta & \sin\theta & -\frac{L}{2} & 0 & -r \end{bmatrix}.$$